Assignment 2

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0.1 Instructions

This problem set is based on Lectures 7–11. The questions are updated as we cover each lecture.

For each question:

1. Understand the economic intuition before using formulas.
2. Be clear about what the model is trying to answer.
3. Use concise reasoning.

1 Lecture 7: Interest-Rate Models

1.1 Question 1

A junior analyst says:

> "Interest rates are uncertain, so we should use a short-rate model for every bond we price."

The analyst is comparing the following securities:

Security	Description
A	5-year Treasury note with fixed annual coupons
B	5-year fixed-rate corporate bond with no embedded options
C	5-year callable agency bond
D	Interest-rate cap
E	Mortgage-backed security

Answer the following:

a. Explain why Securities A and B usually do not require a stochastic short-rate model for time-0 valuation.

b. State the basic valuation logic for a plain fixed-cash-flow bond using discount factors.

c. Explain what a bootstrapped discount curve provides.

d. Identify which securities in the table require future interest-rate dynamics and explain why.

e. Correct the analyst's statement in one or two sentences.

1.2 Question 2

A portfolio manager asks why Lecture 7 spends time modeling the short rate if today's yield curve is already observable.

Answer the following:

a. Explain the difference between a yield curve and a short-rate model.

b. Explain what is meant by saying that a yield curve is a "snapshot."

c. Explain what is meant by saying that a short-rate model is a "scenario generator."

d. Explain why future discount curves $P(t, T)$ matter for securities whose cash flows depend on future rates.

e. Give two examples of portfolio or risk-management questions that cannot be answered well using only today's yield curve.

1.3 Question 3

Consider the basic one-factor short-rate model:

$$dr_t = b(r_t, t)dt + \sigma(r_t, t)dz_t.$$

Answer the following:

a. Explain the role of the drift term.

b. Explain the role of the volatility term.

c. Explain what mean reversion means and why it may be a reasonable assumption for interest rates.

d. Compare the intuition behind constant volatility, level-proportional volatility, and square-root volatility.

e. Explain one practical consequence of choosing a normal model that allows negative rates.

1.4 Question 4

A fixed-income team is deciding between an equilibrium model and an arbitrage-free model.

Answer the following:

a. Explain the main purpose of an equilibrium interest-rate model.

b. Explain the main purpose of an arbitrage-free interest-rate model.

c. Explain why arbitrage-free models are usually preferred for pricing derivatives and bonds with embedded options.

d. In the Hull-White model,

$$dr_t = [\theta(t) - ar_t]dt + \sigma dW_t,$$

explain why $\theta(t)$ is central to fitting today's curve.
e. Explain why a and σ are usually calibrated to interest-rate option prices rather than plain bond prices.

1.5 Question 5

An analyst wants to estimate historical short-rate volatility from weekly data. The analyst observes the following weekly changes in the short rate:

Week	Change in short rate
1	+4 bps
2	-2 bps
3	+6 bps
4	-5 bps
5	+3 bps
6	+1 bp

Week	Change in short rate
7	-4 bps
8	+5 bps

Answer the following:

a. Explain why rate changes, rather than rate levels, are used to estimate volatility in this setting.

b. Describe the steps for estimating weekly historical volatility from these observations.

c. Compute the sample standard deviation of the weekly changes in basis points.

d. Annualize the weekly volatility using $\sqrt{52}$.

e. Explain two limitations of using this historical volatility estimate as an input to a short-rate model.

2 Lecture 8: Bonds with Embedded Options and Option-Adjusted Measures

2.1 Question 6

A 3-year annual-pay corporate bond has face value 100 and coupon rate 5%. The Treasury zero rates are:

Maturity	Zero rate
1 year	3.00%
2 years	4.00%
3 years	5.00%

The bond trades at 99.00.

Answer the following:

a. Write the promised cash flows of the bond.

b. Compute the value of the bond if its promised cash flows were discounted using only the Treasury zero curve.

c. Explain why the market price can be lower than this Treasury-discounted value.

d. Set up the static spread equation for this bond.

e. Use the following trial present values to estimate the static spread and explain why the price-spread relationship is nonlinear:

Spread	Present value
0 bps	100.18
50 bps	98.83
100 bps	97.50

2.2 Question 7

A 3-year annual-pay bond has face value 100 and coupon 7. The risk-neutral probability is $p = 0.5$. The short-rate lattice is:

State	Year 0	Year 1	Year 2
Low-rate state	5%	4%	3%
Middle state		8%	6%
High-rate state			10%

The bond is callable at 101 beginning in year 2.

Answer the following:

a. Compute the terminal payoff at year 3.

b. Compute the year-2 option-free hold values at the 3%, 6%, and 10% nodes.

c. Apply the call rule at year 2. Which nodes are called?

d. Roll the callable values back to year 1.

e. Compute the callable bond value at time 0 and compare it with the option-free value.

2.3 Question 8

A 3-year annual-pay bond has face value 100 and coupon 4. The risk-neutral probability is $p = 0.5$. The short-rate lattice is:

State	Year 0	Year 1	Year 2
Low-rate state	5%	3%	4%
Middle state		9%	8%
High-rate state			12%

The bond is putable at 100 beginning in year 1.

Answer the following:

a. Compute the terminal payoff at year 3.

b. Compute the year-2 option-free hold values.

c. Apply the put rule at year 2. Which nodes are put?

d. Roll the putable values back to year 1 and apply the put rule again.

e. Compute the putable bond value at time 0 and compare it with the option-free value.

2.4 Question 9

An analyst says:

> "A binomial interest-rate tree is determined by the starting short rate, volatility, and time step, so it cannot also fit the full zero curve."

Assume a model uses the following structure:

$$r_{t,j} = a_t e^{(2j-t)\sigma}.$$

Answer the following:

a. Explain what σ controls in this tree.

b. Explain what the time-dependent level parameter a_t controls.

c. Explain why a valuation-grade tree needs one calibration condition for each maturity on the zero curve.

d. Suppose the zero curve has four annual maturities. How many level parameters are needed to fit the four discount factors?

e. Correct the analyst's statement in one or two sentences.

2.5 Question 10

A callable bond is priced with a calibrated 4-period lattice. The model price with OAS equal to 0 is 103.25, while the observed market price is 101.00. The analyst then tries several OAS values:

Trial OAS	Model price
0 bps	103.25
50 bps	101.80
75 bps	101.08
80 bps	100.94

After solving for OAS, the analyst computes effective duration and convexity using a 50 bp parallel curve shock. The model produces:

Scenario	Repriced value
Base curve	101.00
Curve down 50 bps	103.20
Curve up 50 bps	98.90

Answer the following:

a. Explain what OAS is adding to the lattice.

b. Estimate the OAS from the trial values.

c. Explain why the OAS is positive in this example.

d. Compute effective duration.

e. Compute effective convexity and explain what these measures capture for a callable bond.

3 Lecture 9: Interest-Rate Futures Contracts

3.1 Question 11

A trader buys one Treasury futures contract at 100-08. The contract size is $100,000, and one full point is worth $1,000. The next three daily settlement prices are:

Day	Settlement price
0	100-08
1	100-20
2	99-28
3	100-04

Answer the following:

a. Explain what it means to be long a futures contract.

b. Convert each settlement price into decimal form.

c. Compute the daily mark-to-market cash flow to the long position.

d. Compute the cumulative cash flow over the three days.

e. Explain why the trader does not pay the full notional value when entering the futures position.

3.2 Question 12

A trader buys one Three-Month SOFR futures contract at 95.36. The next settlement price is 95.42. Each 1-basis-point price move is worth approximately $25 per contract.

Answer the following:

a. Compute the initial implied SOFR rate.

b. Compute the new implied SOFR rate.

c. Compute the price change in basis points.

d. Compute the dollar gain or loss for the long futures position.

e. Explain why a higher SOFR futures price corresponds to a lower implied SOFR rate.

3.3 Question 13

A Treasury futures contract has settlement price 96.00. Three eligible deliverable bonds have the following cash prices and conversion factors:

Bond	Cash price	Conversion factor
A	101.40	1.0400
B	98.20	1.0150
C	94.10	0.9700

Ignore accrued interest and financing.

Answer the following:

a. Explain what the cheapest-to-deliver bond is.

b. Compute the invoice amount before accrued interest for each bond.

c. Compute the net delivery cost for each bond.

d. Identify the cheapest-to-deliver bond.

e. Explain why the lowest cash price alone does not necessarily identify the cheapest-to-deliver bond.

3.4 Question 14

A bond sells for 100, pays a 5 percent annual coupon, and the futures contract settles in six months. The financing rate is 7 percent per year.

Answer the following:

a. Compute the theoretical futures price using:

$$F = P[1 + t(r - c)].$$

b. Explain whether the futures price should be above or below the cash price.

c. If the market futures price is 103, identify the arbitrage trade and compute the approximate arbitrage profit.

d. If the market futures price is 99, identify the arbitrage trade and compute the approximate arbitrage profit.

e. Explain why arbitrage pressure pushes the futures price toward the theoretical price.

3.5 Question 15

A portfolio manager wants to hedge a long bond portfolio using Treasury futures. The portfolio has DV01 of \$27,840. The relevant futures contract has a cheapest-to-deliver bond with:

- modified duration = 6.5;
- CTD clean price = 98.50;
- deliverable face amount = \$100,000;
- conversion factor = 0.9200.

Answer the following:

a. Explain why Treasury futures do not have a true Macaulay or modified duration.

b. Compute the CTD bond's market value for the deliverable face amount.

c. Compute the CTD bond's DV01.

d. Compute the approximate futures DV01.

e. Compute the number of futures contracts needed for a full DV01 hedge and state whether the manager should buy or sell futures.

4 Lecture 10: Credit Spread Exposure and Credit Risk Modeling

4.1 Question 16

A corporate bond has market value 4,000,000, spread duration of 4.5, and a credit spread of 180 basis points.

Answer the following:

a. Explain what spread duration measures.
b. Estimate the bond's percentage price change if its spread widens by 35 basis points.
c. Estimate the dollar price change.
d. Compute duration times spread (DTS).

e. Explain why DTS can be more useful than spread duration alone when comparing bonds with different credit quality.

4.2 Question 17

A corporate bond has analytical duration of 6.0. A regression of its weekly return on the weekly Treasury return produces a slope coefficient of 0.65.

Answer the following:

a. Distinguish analytical duration from empirical duration.
b. Compute the bond's empirical duration using the duration-multiplier approach.
c. Estimate the percentage price change associated with a 40-basis-point increase in Treasury yields using analytical duration.
d. Repeat the estimate using empirical duration.
e. Explain why the empirical estimate may be smaller for a lower-rated corporate bond.

4.3 Question 18

In a one-period structural credit model, a firm has asset value of $120 million today. At the debt maturity date, it must repay debt with face value of $100 million.

Answer the following:

a. State the condition under which the firm defaults at maturity.
b. Describe the bondholders' payoff at maturity.
c. Describe the shareholders' payoff at maturity and explain its option interpretation.
d. Explain how an increase in asset volatility affects default risk, holding other inputs constant.
e. Give two practical limitations of structural credit models.

4.4 Question 19

Two portfolios each contain 100 equal-sized issuers. Every issuer has a one-year default probability of 2%. Defaults are independent in Portfolio A but strongly exposed to a common recession factor in Portfolio B.

Answer the following:

a. Compute the expected number of defaults in each portfolio.
b. Explain whether the expected number of defaults differs across the portfolios.
c. Identify which portfolio has greater tail risk and explain why.
d. Explain why pairwise correlation may not fully describe credit-portfolio risk.

e. Explain what a copula adds to the marginal default distributions.

4.5 Question 20

A reduced-form model assumes a constant annual default intensity of $\lambda = 0.02$ and a loss given default of 60%.

Answer the following:

a. Interpret the annual default intensity in words.
b. Compute the one-year survival probability and default probability.
c. Compute the five-year survival probability and cumulative default probability.
d. For 100 similar issuers that survive to the start of the year, compute the expected number of defaults over the next year.
e. Using spread $\approx \lambda \times \text{LGD}$, estimate the credit spread in basis points and state one reason why an observed market spread may differ.

5 Lecture 11: Interest-Rate Options

5.1 Question 21

A trader buys one 110 put on a 10-year Treasury futures contract for a premium of 0-40. Treasury futures option premiums are quoted in 64ths of a point, and one full point is worth $1,000 per contract.

Answer the following:

a. Convert the premium quote 0-40 into decimal price points.

b. Compute the dollar premium paid.

c. Compute the put's break-even futures price at expiration in both decimal and Treasury fractional notation.

d. Compute the put's payoff and net profit if the futures price at expiration is 107-16.

e. Compute the payoff and net profit if the futures price at expiration is 112-00, and explain whether the put should be exercised.

5.2 Question 22

European options on a Treasury futures contract have the following terms:

- current futures price $F_0 = 111$;
- strike $K = 110$;
- time to expiration $T = 0.25$ years;
- continuously compounded risk-free rate $r = 4.5\%$; and
- call premium $c = 2.2823$ points.

Answer the following:

a. State the put-call parity relationship for European futures options.

b. Compute the present-value factor e^{-rT}.

c. Use parity to compute the fair put premium.

d. Suppose the market put premium is 1.5000 points. Determine whether the call or put side of parity is relatively expensive and calculate the mispricing per contract in dollars.

e. State the arbitrage positions and calculate the locked-in time-0 profit per contract, ignoring transaction costs and margin effects.

5.3 Question 23

Consider a European option on a zero-coupon bond with:

- bond price $S_0 = 95$;
- strike $K = 94$;
- option maturity $T = 0.75$ years;
- continuously compounded risk-free rate $r = 3\%$; and
- annualized bond-price volatility $\sigma = 12\%$.

Use the Black-Scholes formulas for a non-coupon-paying asset.

Answer the following:

a. Compute $\sigma\sqrt{T}$.

b. Compute d_1 and d_2.

c. Using $N(d_1) = 0.6444$ and $N(d_2) = 0.6050$, compute the call value.

d. Using $N(-d) = 1 - N(d)$, compute the put value.

e. Verify numerically that put-call parity holds.

5.4 Question 24

A European call is priced with a two-step Cox-Ross-Rubinstein tree. The inputs are:

- current underlying price $S_0 = 100$;
- strike $K = 100$;
- option maturity $T = 0.5$ years;
- annualized volatility $\sigma = 20\%$;
- continuously compounded risk-free rate $r = 4\%$; and
- number of steps $N = 2$.

Answer the following:

a. Compute Δt, $u = e^{\sigma\sqrt{\Delta t}}$, and $d = 1/u$.

b. Compute the risk-neutral up probability $q = (e^{r\Delta t} - d)/(u - d)$.

c. Compute the three possible underlying prices at expiration.

d. Compute the call payoff at each terminal node.

e. Use backward induction to compute the call value at time 0.

5.5 Question 25

An at-the-money European call on a zero-coupon bond has $S_0 = K = 100$, $T = 0.5$ years, and $r = 3\%$. Its observed market price is 3.4000. Trial Black-Scholes values are:

Trial volatility	Model call value
6%	2.5281
8%	3.0623
10%	3.6065

Answer the following:

a. Explain numerically why the implied volatility lies between 8% and 10%.

b. Estimate the implied volatility by linear interpolation between the 8% and 10% trial values.

c. Calculate the pricing error if 8% volatility were used.

d. Calculate the pricing error if 10% volatility were used.

e. If the market price rises to 3.5000 with all other inputs unchanged, estimate the new implied volatility using the same interpolation interval and explain why it increases.